FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Manual Vacuum Aspiration (MVA) Kit

Clinical skill | Week 2 · Labour, delivery & the puerperium | Dr Nsingo, Dr Kavuma

MVA uses a hand-held 60 mL aspirator that creates a vacuum of about 60 cm Hg, transmitted through a plastic cannula, to empty the uterus under paracervical block. It is the WHO-preferred method for uterine evacuation below about 12–14 weeks' uterine size, can be done by trained midwives at HC III/IV, and replaces sharp curettage.

Must know before the ward round

1. The Ipas-type aspirator holds 60 mL and generates a vacuum of about 60–66 cm Hg (24–26 inches Hg). Cannulae come in 4–12 mm; pick one about equal to the uterine size in weeks, adjusted to how far the os is open.
2. Indications: incomplete, inevitable and missed abortion; induced abortion where legal; endometrial biopsy; evacuation of a small molar pregnancy. Uterine size up to about 12 weeks (WHO allows vacuum aspiration up to 14 weeks with skill and the right cannula).
3. Before you start: vitals stable (or resuscitation under way), Hb and Rhesus group known, IV antibiotics already running if septic, ectopic excluded, and a bimanual exam for uterine SIZE and POSITION.
4. Septic abortion is not a contraindication: start IV antibiotics first, then do MVA within hours. Suspected ectopic or perforation IS a reason not to aspirate.
5. Paracervical block: 1% lidocaine (lignocaine) up to 20 mL (maximum 4.5 mg/kg plain). Aspirate before every injection. Give ibuprofen 400–800 mg orally 30–60 minutes before.
6. Closed cervix (missed abortion): prime with misoprostol 400 µg sublingually 1 hour or vaginally 3 hours before MVA.
7. Signs of completion: red or pink foam, no more tissue in the cannula, a gritty feel, and the uterus gripping the cannula.
8. Never push the plunger while the cannula is still in the uterus. Detach first, so you never push air or contents back into the uterus.
9. Perforation signs: the cannula passes further than the uterine size allows, sudden pain, loss of resistance, or fat or bowel in the aspirate. Stop, and observe or do a laparotomy.
10. Always examine the aspirated tissue: villi and a sac confirm an intrauterine pregnancy. If there is no tissue, think incomplete evacuation, ectopic or mole.
11. After MVA: pulse, BP, bleeding and pain every 15 minutes for 30–60 minutes; anti-D within 72 h if Rh-negative; contraception (IUD can go in at once unless septic).
12. Processing: keep wet, then clean with detergent and water, then high-level disinfection (boil 20 min, or 0.5% chlorine or 2% glutaraldehyde for 20 min) or sterilisation. Prepare 0.5% chlorine from JIK (3.5%) as 1 part JIK to 6 parts water.

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1. Purpose and Indications

- **Manual vacuum aspiration (MVA)** empties the uterus by suction produced with a **hand-held syringe-like aspirator**. It needs no electricity, no theatre and no general anaesthesia.
- WHO and Uganda MoH recommend **vacuum aspiration** (manual or electric) as the surgical method of choice for uterine evacuation in the first trimester. **Sharp curettage (D&C) should be replaced**, because MVA causes less bleeding, pain, perforation and scarring.
- In Uganda, MVA is the backbone of **post-abortion care (PAC)** at HC III, HC IV and hospitals. **Trained midwives and clinical officers perform it**, which makes it one of the most important lifesaving skills you will see on your rotation.
- The procedure is short (5–10 minutes of aspiration), done under **paracervical block + NSAID + verbal support**.
- **The one thing never to forget:** MVA is not finished when the aspiration stops. You must **examine the tissue**, give **contraception**, and **process the instruments** so the next woman is safe.

Indications

1. **Incomplete abortion** (the commonest use in Uganda), with uterine size up to about 12 weeks. WHO allows vacuum aspiration up to 14 weeks with an experienced provider.
2. **Inevitable abortion.**
3. **Missed abortion / early pregnancy loss** (with or without cervical priming).
4. **Induced abortion** within the legal indications.
5. **Septic abortion**, **after IV antibiotics have started**. Infection is **not** a contraindication; evacuation is the treatment.
6. **Molar pregnancy** in a small uterus. Electric vacuum is preferred for a large mole, with oxytocin running and blood ready.
7. **Endometrial biopsy** (e.g. with a 4 mm cannula) in abnormal uterine bleeding.

2. Contraindications and Precautions

Situation	What to do
Uterine size larger than about 12–14 weeks	Beyond MVA capacity for most providers. Use **misoprostol 400 µg every 3 hours**, then remove residual tissue; or D&E by a skilled provider.
Suspected ectopic pregnancy (empty uterus, adnexal mass)	Do not do MVA. Work up for ectopic.
Haemodynamic instability	**Resuscitate first**. Remove POC from the os; do MVA as soon as she is stabilising. Do not delay indefinitely, because the retained tissue is the cause of the bleeding.
Septic abortion	**Start IV antibiotics first**, then evacuate promptly
Suspected uterine perforation or bowel injury	Do not aspirate blindly. Laparotomy or specialist review.
Known coagulopathy	Correct it, have blood ready, do it at a hospital
Distorted cavity (fibroids, uterine anomaly) or **previous difficult evacuation**	Experienced provider; ultrasound guidance if available
Allergy to lidocaine (rare)	Use an NSAID and sedation-based analgesia
Acute cervicitis	Treat it, but do not delay an evacuation that is needed
Asthma, peptic ulcer, renal disease	Avoid NSAIDs; use paracetamol

Situation	What to do
Previous CS, myomectomy, fibroids	Scarred or distorted uterus: higher perforation risk; experienced provider

> RED FLAG
> **Stop and rethink** if there is **one-sided pain, shoulder-tip pain, a closed os with an empty uterus, or peritonism**: this may be an ectopic or a perforation, and MVA is the wrong operation.

3. Relevant Anatomy and Normal Values

How the vacuum works

- When the valves are closed and the plunger is pulled back until its arms lock, the aspirator cylinder holds a **negative pressure of about 60–66 cm Hg**. That is close to the vacuum produced by an electric pump.
- Pressing the valve buttons open **transfers the vacuum through the cannula** to the uterine cavity.
- The soft, loosely attached products of conception (POC) and decidua are sucked into the cannula aperture, while the firmer myometrium is not.
- As the uterus empties, it **contracts down** on the cannula. That is why you feel it "grip" and why bleeding decreases.

Why MVA is safer than sharp curettage

- The plastic cannula is **blunt and flexible**, so it is less likely to perforate the uterus and does not scrape the basal layer of the endometrium. Stripping the basal layer is what causes **Asherman syndrome** (intrauterine adhesions).
- It needs little or no cervical dilation for incomplete abortion, because the os is usually already open.
- Local anaesthesia is enough, so there is no general-anaesthetic risk and no theatre queue.

Pain physiology (why the paracervical block works)

- The uterus and cervix are supplied by the **uterovaginal (Frankenhäuser) plexus**, which lies lateral to the cervix at the cervicovaginal junction.
- Lidocaine injected at **4 and 8 o'clock** (± 2 and 10 o'clock) bathes this plexus and blocks sensation from the cervix and the lower uterus.
- The block does not abolish **fundal cramping**, which travels with sympathetic fibres (T10–L1). That is why an NSAID (which cuts prostaglandin-driven cramps) and **"vocal local"** (calm talking) are added.

Uterine position and the uterocervical angle

- Know whether the uterus is **anteverted or retroverted** before any instrument goes in. **Most perforations happen when the provider does not know the uterus is retroverted** and pushes the cannula anteriorly.
- Tenaculum traction **straightens the uterocervical angle** and protects against perforation more than any other manoeuvre.
- The **cervicovaginal junction** (where smooth cervical epithelium meets the vaginal rugae) is the landmark for the block.

Sizing the uterus on bimanual (commonly taught comparisons; practice varies)

Uterine size	What you feel
Non-pregnant	Small, firm, like a pear

Uterine size	What you feel
About 6 weeks	Slightly enlarged, soft (like a hen's egg to a small orange)
About 8 weeks	Like an orange
About 10 weeks	Like a grapefruit; fills much of the pelvis
About 12 weeks	Fills the pelvis; fundus **just palpable above the symphysis pubis**
About 14 weeks or more	Easily palpable abdominally: beyond routine MVA; use misoprostol

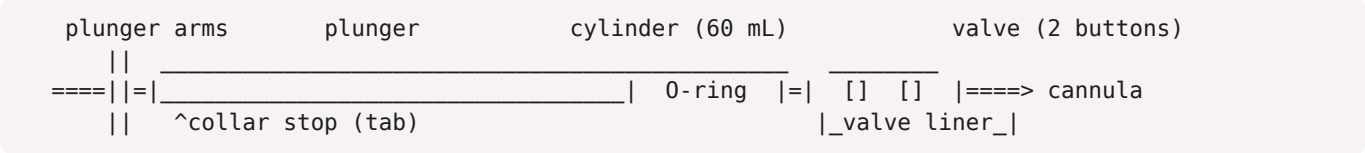
A full bladder, obesity, fibroids or a retroverted uterus make sizing less accurate. Always compare the **uterine size with the dates**: smaller suggests incomplete or missed abortion; larger suggests mole, twins, fibroids or wrong dates. The fundus lies roughly at **9–10 cm** from the os in a 9-week uterus.

Key numbers

Item	Value
Aspirator capacity	**60 mL**
Vacuum	About **60–66 cm Hg** (24–26 in Hg)
Cannula sizes	**4–12 mm**; size in mm ≈ uterine size in weeks
Lidocaine 1%	**10 mg/mL**; max **4.5 mg/kg** plain
Shock index (pulse ÷ SBP)	**0.9 or more abnormal, 1.4 or more critical**
Bedside clotting test	No clot by **7 minutes** = abnormal

4. Equipment and Preparation

The MVA kit (Ipas-type MVA Plus aspirator)



Part	Function / what to check
Cylinder (barrel)	Holds 60 mL of aspirate. Check for cracks and clarity.
Plunger with plunger arms	Pulling it back creates the vacuum. The arms must **lock over the cylinder rim**, or the plunger may spring back into the cylinder.
O-ring	Seals the plunger. **Lubricate with a thin film of silicone** so it keeps the vacuum.
Collar stop (with retainer tab)	Keeps the plunger from coming out of the cylinder
Valve with 2 buttons + valve liner + cap	Closed buttons hold the vacuum; pressing them releases it into the cannula. The liner must be seated correctly.
Cannulae (e.g. Ipas EasyGrip)	Sizes **4, 5, 6, 7, 8, 9, 10, 12 mm**, colour-coded. **Dots mark centimetres from the tip**, so you can gauge depth. The aperture(s) near the tip are where tissue enters.
Adapters	Connect cannulae from other makers to the aspirator
Silicone lubricant	For the O-ring

Other instruments and supplies

Item	Use
Speculum (Cusco's bivalve or Sims)	Expose the cervix
Single-tooth tenaculum (or vulsellum / sponge forceps)	Stabilise the cervix and apply traction to straighten the canal
Sponge-holding forceps	Remove POC from the os; hold swabs for antiseptic
Tapered dilators (Denniston plastic or Pratt/Hegar metal)	Dilate a closed or partly open cervix
Antiseptic (povidone-iodine or chlorhexidine) and gallipot, gauze	Cervical and vaginal preparation
Lidocaine 1% (plain), **10–20 mL syringe**, long needle (22 G spinal or needle extender)	Paracervical block
Strainer, clear bowl, water or vinegar, light source	Examining the tissue
PPE: gloves, apron or gown, eye protection	Infection prevention
Emergency tray: oxytocin, misoprostol, ergometrine, atropine, IV fluids, oxygen, bag-valve mask, diazepam	Complications
Contraceptives: IUD insertion kit, implants, DMPA	Immediate post-abortion family planning
Processing supplies: buckets, detergent, soft brush, 0.5% chlorine (JIK), boiler or glutaraldehyde, covered storage container	Instrument processing

Choosing the cannula

Uterine size (weeks)	Suggested cannula (mm)
4–6	4–7
7–9	5–10
9–12	8–12

Rules of thumb:

- **Cannula size ≈ weeks of uterine size.** For **incomplete abortion**, choose the **largest cannula that passes the open os comfortably**.
- **Too small** a cannula clogs and leaves tissue behind.
- **Too large** a cannula tears the cervix or needs more dilation.

Pre-procedure assessment

For a skill like MVA, the "workup" is the **pre-procedure assessment** (is MVA the right operation, for this woman, now?), the **preparation** (consent, analgesia, kit check) and the **post-procedure checks**. On a Ugandan gynae ward the MVA is often done within an hour of admission, so this has to be quick, but no step can be skipped.

First 5 minutes

1. **Is she stable?** Pulse, BP, respiratory rate, temperature, SpO_2, level of consciousness. Calculate the **shock index** (pulse ÷ systolic BP): **0.9 or more abnormal, 1.4 or more critical**.
2. **If shocked:** call for help, two large-bore cannulae (16–18 G), bloods (Hb, group and crossmatch), **1 L crystalloid quickly**, then reassess. **Remove POC from the os with sponge forceps** at the bedside; this may stop the bleeding and reverse cervical shock. MVA follows **as soon as she is stabilising**, because the retained tissue is the cause of the bleeding.

3. **If febrile (38 °C or more), with foul discharge or a tender uterus:** give the first doses of **IV ampicillin + gentamicin + metronidazole** now; MVA follows within hours.
4. **Heavy bleeding and the MVA room not ready?** Uterotonic: **misoprostol 400 µg oral/sublingual** or **ergometrine 0.2 mg IM** (not if hypertensive).
5. **Stop and rethink** if there is one-sided pain, shoulder-tip pain, a closed os with an empty uterus, or peritonism.

Focused history: what to ask and why

Ask	Why (what changes the plan)
LMP, uterine size, any scan	Expected uterine size and cannula size; whether MVA is appropriate; a prior scan showing an IUP makes an ectopic unlikely
Bleeding volume, dizziness, fainting	Resuscitate first; urgency of the procedure
Tissue passed; interference; fever; foul discharge	Septic: IV antibiotics before MVA. Instrumentation: higher risk of perforation and bowel injury already present, so examine the abdomen carefully first
Pain pattern (midline cramps vs unilateral, shoulder-tip)	Unilateral pain or collapse = ectopic: do not aspirate
Previous CS, myomectomy, fibroids, uterine anomaly, previous difficult evacuation	Distorted or scarred uterus: higher perforation risk; experienced provider, ultrasound guidance if available
Bleeding disorder, anticoagulants	Correct coagulopathy; do it in hospital with blood ready
Allergies (lidocaine, iodine, NSAIDs, penicillin)	Analgesia, antiseptic and antibiotic choice
Asthma, peptic ulcer, renal disease	Avoid NSAIDs; use paracetamol
Blood group / Rhesus, HIV status	Anti-D; opt-out HIV testing and linkage
Contraceptive intention	Have the IUD kit or implant ready so it can be given in the same visit
Last meal	Rarely matters for MVA under local anaesthesia, but record it in case you need to convert to theatre

Focused examination (Hutchison order)

1. **Introduce yourself, consent, chaperone, hand hygiene**; position and expose only as needed.
2. **General inspection and vital signs:** distress, pallor, temperature, pulse, BP, RR, shock index.
3. **Abdomen:** inspection for distension; palpation for tenderness, **guarding, rebound or distension** (perforation, abscess or ectopic: do not proceed without senior review); is the uterus palpable above the symphysis?
4. **Speculum:** os open or closed and how wide; POC in the os (remove them); offensive discharge; tears, burns, tenaculum marks or foreign bodies.
5. **Bimanual (the essential safety step), BEFORE the procedure:**
   - **Uterine size in weeks** (predicts cannula size and expected depth).
   - **Position: anteverted or retroverted.** An unrecognised retroverted uterus is the classic set-up for perforation.
   - Consistency, tenderness, cervical motion tenderness, adnexal masses.
6. Thank her, hand hygiene, summarise and explain the plan.

Bedside tests

Test	Why
HemoCue / bedside Hb	Baseline. Severe anaemia (under 7 g/dL) or shock: transfuse and have blood ready.

Test	Why
Urine pregnancy test	Confirms pregnancy if not already known.
Bedside clotting test (2 mL in a plain glass tube; no clot by 7 minutes = abnormal)	Septic or long-retained pregnancy: suspect DIC before instrumenting.
mRDT	Fever: co-existing malaria, but it never excuses you from treating sepsis.
Point-of-care ultrasound (if available and you are trained)	Retained products vs empty uterus; free fluid; confirms size before and completeness after.
Aspirator vacuum test	Close the valves, pull the plunger until the arms lock, wait a few minutes, then open the valves: a rush of air means it holds the vacuum. Recharge before use.

Laboratory and imaging

Test	When	Expected / abnormal finding	How it changes management
Hb / FBC	Everyone (Hb at HC III; FBC at HC IV and above)	Anaemia; raised WBC in sepsis; low platelets	Transfuse; supports sepsis; check clotting before MVA
Blood group and Rhesus D	Everyone before or at the procedure (HC IV and above)	Rh-negative	**Anti-D within 72 h** after surgical evacuation (UK 250 IU before 20 weeks; the Ugandan vial usually stocked is 1,500 IU) Verify *anti-D presentation stocked in Uganda*
Crossmatch	Heavy bleeding, Hb under 7 g/dL, suspected mole or coagulopathy (HC IV and above)	–	Blood ready in the room before you start
HIV and syphilis tests	Everyone (all levels; opt-out)	Positive	Linkage to ART; treat syphilis
Pelvic ultrasound	Uncertain diagnosis, closed os, suspected ectopic, mole or perforation; after MVA if no villi seen. Abdominal scan at most HC IVs; TVUS mainly at referral hospitals	Retained products; empty uterus ± adnexal mass or free fluid; snowstorm (mole)	Confirms MVA is appropriate; an empty uterus with no villi → ectopic pathway; mole → electric vacuum at a hospital
Serum β-hCG	No villi in the aspirate, suspected ectopic or mole (mainly referral hospitals and private labs)	Rising or plateauing hCG after MVA	Ectopic or failed procedure; very high hCG = mole
Renal function, clotting screen	Septic abortion, oliguria, abnormal bedside clotting (regional/national referral)	Raised creatinine; prolonged PT/APTT, low fibrinogen	Gentamicin dose; blood products before evacuation
Histology	Where available, especially if vesicles or no villi seen	Villi (confirms IUP); hydropic villi (mole)	Mole → hCG follow-up

Consent, analgesia and positioning

- **Consent:** what MVA is; that it takes 5–10 minutes of aspiration; pain relief (block + NSAID + talking); risks (**bleeding, infection, incomplete evacuation, perforation under 1 in 1,000**, cervical injury); alternatives (misoprostol, expectant); contraception offered; questions answered; written consent and a chaperone.
- **Analgesia: ibuprofen 400–800 mg orally 30–60 minutes before**, ± paracetamol 1 g. Consider **diazepam 5–10 mg orally** for anxiety (monitor her if you give it).

- **Antibiotics: IV antibiotics** if septic. Otherwise follow local policy. WHO recommends **prophylactic antibiotics before surgical abortion**, e.g. **doxycycline 200 mg orally** or **metronidazole 500 mg orally** once before the procedure.
- **Cervical priming** for a closed cervix (missed abortion): **misoprostol 400 µg sublingually 1 hour (or vaginally 3 hours) before**.
- **Bladder emptied**; **lithotomy position**; perineum exposed, the rest of her covered.

The pre-procedure checklist (say it aloud before you start)

1. **Indication confirmed:** e.g. incomplete abortion, uterus under 12 weeks, ectopic excluded.
2. **Prerequisites met:** vitals stable or resuscitation running; Hb and group known; IV antibiotics already given if septic; bladder emptied; **ibuprofen 400–800 mg given 30–60 minutes ago**; prophylactic antibiotic per local policy (e.g. doxycycline 200 mg or metronidazole 500 mg orally once).
3. **Consent** obtained as above.
4. **Equipment check:** aspirator assembled, O-ring lubricated, **valves closed, plunger arms locked, vacuum tested**; cannulae of the expected size and one size either side; dilators, tenaculum, speculum, sponge forceps; antiseptic; lidocaine 1% drawn up (**maximum 4.5 mg/kg**); strainer, clear dish, water or vinegar and light; emergency drugs (oxytocin, misoprostol, ergometrine, atropine, diazepam), IV fluids, oxygen; FP supplies; processing buckets ready.

5. Procedure Step by Step

A. Before the procedure

1. **Counsel and consent.** Explain what MVA is, how long it takes, the pain relief, the risks (bleeding, infection, incomplete evacuation, perforation under 1 in 1,000), and the alternatives (misoprostol). Offer contraception. Obtain written consent.
2. **Assess:** history, vitals, Hb, blood group and Rhesus, HIV test (opt-out), bimanual exam for **uterine size and position**.
3. **Resuscitate** if needed. **Antibiotics:** IV if septic; otherwise prophylaxis per local policy (doxycycline 200 mg or metronidazole 500 mg orally once).
4. **Pain medication: ibuprofen 400–800 mg orally 30–60 minutes before**, ± paracetamol 1 g; consider diazepam 5–10 mg orally for anxiety.
5. **Ask her to empty her bladder.**
6. **Prepare and check the aspirator:**
   - It should be assembled.
   - The O-ring should be lubricated.
   - The plunger should be fully inserted.
   - **Close the valves** (push the buttons down and forward until they lock).
   - **Pull the plunger back until the arms lock** over the cylinder rim.
   - **Test the vacuum** before starting. Leave it charged for a few minutes, then open the valves: you should hear a rush of air.
   - **Recharge it** before use.
7. **Arrange** the sterile or HLD instruments on a tray, the cannulae in their sizes, and the emergency drugs.

B. Cervical preparation and paracervical block

8. **Wash your hands, and put on PPE and gloves.** Use a **no-touch technique**: never touch the part of the cannula that enters the uterus.

9. With her in the **lithotomy position**, insert the **speculum** and visualise the cervix. **Remove any POC from the os** with sponge forceps.
10. **Clean the cervix and os** with antiseptic-soaked gauze on sponge forceps, in a spiral moving outwards, 2–3 times.
11. **Paracervical block:**
    - Draw up **lidocaine 1%, up to 20 mL** (200 mg; do not exceed **4.5 mg/kg** without adrenaline).
    - Inject **1–2 mL at the tenaculum site** (12 o'clock on the anterior lip), then apply the **tenaculum**.
    - With gentle traction, find the **cervicovaginal junction** (where smooth cervical epithelium meets the vaginal rugae).
    - At **4 and 8 o'clock** (± 2 and 10 o'clock), insert the needle **1.5–3 cm deep**. **Aspirate first**, to make sure you are not in a vessel. Then inject **3–5 mL at each site**, moving the needle slowly as you inject.
    - **Wait 2–3 minutes** for the block to work. Use this time to talk to her.

DRUG DOSING

Lidocaine 1% = 10 mg/mL. 20 mL = 200 mg, which is safe for women of about 45 kg or more (4.5 mg/kg). For a 40 kg adolescent, the maximum is **18 mL**. Always aspirate before injecting. Toxicity begins with **perioral numbness and tinnitus**; stop immediately.

Which maximum? Sources differ: **4.5 mg/kg (max 300 mg)** for plain lidocaine (WHO abortion care and US labelling, used here), but the BNF and this site's drug handbook and perineal-repair notes teach a more cautious **3 mg/kg**. In a small or adolescent woman, use the lower figure and dilute to 0.5% if more volume is needed. Verify *which maximum the unit uses*

C. Dilation (only if needed)

12. Hold **steady traction** on the tenaculum to straighten the canal.
13. If the os will not admit the chosen cannula, **dilate progressively** with tapered dilators (or with successively larger cannulae), rotating gently. **Never force.** In incomplete abortion, the os is often already open enough.

D. Aspiration

14. **Insert the cannula** through the os with a gentle rotating motion until it **just touches the fundus**, then **withdraw it about 1 cm**. Note the depth from the dots.
    - Do **not** use the cannula to "sound" the uterus vigorously.
    - **Do not attach the aspirator before insertion.**
15. **Attach the charged aspirator** to the cannula. Hold the tenaculum and the cannula steady so the cannula does not advance as you attach.
16. **Release the pinch valves** by pressing both buttons. The vacuum transfers to the uterus, and **blood and tissue** appear in the cannula and cylinder.
17. **Evacuate the uterus:**
    - **Gently and slowly rotate the cannula 180° in each direction** while moving it **in and out** between the fundus and the internal os.
    - Do not withdraw the aperture past the os, or you lose the vacuum.
    - Hold the aspirator by the cylinder, not by the plunger arms.
18. **Recognise completion:**
    - **red or pink foam**, and no more tissue entering the cannula;
    - a **gritty sensation** as the cannula passes over the empty endometrium;
    - the **uterus contracts and grips the cannula**;
    - she reports more **cramping**.

19. When complete or full:
    - **close the valves**;
    - **detach the aspirator from the cannula** (or withdraw both together without pushing the plunger);
    - empty the aspirator into the strainer by opening the valves and pushing the plunger.
    - **Never push the plunger while the aspirator is connected to a cannula in the uterus.**
    - If more tissue is suspected, recharge and repeat, or use a second aspirator.

E. After aspiration

20. **Inspect the tissue:**
    - Empty it into a **strainer**, rinse it, then float it in **clear water or vinegar in a clear dish, backlit by a light**.
    - Look for **villi** (white, fluffy, frond-like), **decidua** and the **gestational sac / membranes**. From about 9 weeks you may see fetal parts.
    - Check that the amount matches the gestational age.
    - Send tissue for histology where available.
21. Do any **concurrent procedures**:
    - **IUD insertion** (if she chose it and there is no sepsis);
    - repair of a bleeding tenaculum site (pressure or a suture).
22. **Remove the tenaculum**, check the cervix for bleeding, and remove the speculum.
23. **Immediately start instrument processing** (soak and keep wet, step F).
24. **Recovery:**
    - Monitor **pulse, BP, bleeding and pain every 15 minutes for 30–60 minutes**.
    - Give analgesia.
    - Give **anti-D 250 IU IM** if Rh-negative (where available).
    - Start or confirm **contraception**.
25. **Discharge advice:**
    - Expect light bleeding and cramps for up to 1–2 weeks.
    - **Return with danger signs:** fever, heavy bleeding (1 pad or more per hour), severe or increasing pain, foul discharge, fainting.
    - Fertility returns within 2 weeks.
    - Follow-up visit in 1–2 weeks.

F. Instrument processing (decontamination, cleaning, HLD, sterilisation)

Why it matters: the cannula enters the uterine cavity, so poorly processed instruments transmit **HIV, hepatitis B and C** and bacteria. Processing follows the Uganda IPC sequence.

Step	How	Key points
1. Immediately after use: keep wet / decontaminate	Rinse by drawing clean water through the cannula and aspirator. Then **soak all items in 0.5% chlorine for 10 minutes** (Uganda IPC practice), or at least keep them submerged in water.	**Never let blood and tissue dry on the instruments.** Dried material cannot be cleaned out.
2. Clean (the most important step)	**Disassemble the aspirator** (valve, liner, plunger, O-ring). Wash in **warm water with liquid detergent**, using a **soft brush**; flush the cannulae. Rinse in clean water.	HLD and sterilisation **fail if the item is not clean first**. Do not use abrasive powders; they scratch plastic.

Step	How	Key points
3a. High-level disinfection (HLD)	**Boiling for 20 minutes** (timed from rolling boil; items fully submerged), **or** soak in **0.5% chlorine for 20 minutes** (Ipas; general IPC manuals use 0.1% chlorine made with boiled water), **or** in **2% glutaraldehyde (e.g. Cidex) for 20 minutes**	After chemical HLD, **rinse with boiled (sterile) water**.
3b. Sterilisation	**Steam autoclave** (121 °C, 106 kPa, **30 minutes** wrapped / 20 minutes unwrapped) for items rated for it (e.g. the aspirator and metal instruments), **or glutaraldehyde 2% for 8–10 hours**.	**Follow the manufacturer's instructions:** many plastic cannulae **must not be autoclaved** because they deform.
4. Dry and store	Air-dry, store in an **HLD or sterile covered container**, labelled with the date	Reassemble the aspirator, **lubricate the O-ring**, and **test the vacuum** before the next use

UGANDA CONTEXT

Making 0.5% chlorine from JIK (3.5% sodium hypochlorite). Parts of water = (% concentrate ÷ % desired) − 1 = (3.5 ÷ 0.5) − 1 = **6**. So mix **1 part JIK with 6 parts clean water**, for example 1 cup of JIK in 6 cups of water. Make it fresh **daily**, or sooner if it becomes cloudy or bloody. Chlorine **corrodes metal**, so do not soak metal instruments for longer than the recommended time.

NOTE

The **Spaulding classification** divides equipment into *critical* items (enter sterile tissue: sterilise), *semi-critical* items (contact mucous membranes: HLD at least) and *non-critical* items (intact skin: clean). The MVA cannula passes through the non-sterile cervix into the uterus. WHO accepts **HLD or sterilisation**, while some hospitals sterilise all intrauterine instruments.

6. Findings and Interpretation

The aspirated tissue

Finding	Interpretation	Action
Villi (white, fluffy, frond-like), decidua, sac; amount matches the gestation	Intrauterine pregnancy evacuated	Routine aftercare
No villi, no sac	Incomplete evacuation, POC already passed (complete abortion), or **ectopic pregnancy**	Vitals and abdomen, **ultrasound**, serum hCG repeated at 48 h; ectopic safety-net
Grape-like vesicles	**Molar pregnancy**	Histology; hCG follow-up
Yellow fat, omentum or bowel	**Perforation**	Stop; perforation pathway (section 8)
Fetal parts	Seen from about 9 weeks	Confirms IUP

Signs of completion

Sign	Meaning
Red or pink foam, no more tissue	Cavity empty; air and blood only
Gritty sensation	Cannula moving over empty endometrium
Uterus grips the cannula	Myometrium contracting as the cavity empties

Sign	Meaning
Increased cramping	Uterine contraction

Post-procedure checks

Check	Normal	Worry if
Tissue (inspect it)	Villi (white, fluffy, frond-like), decidua, sac; amount matches the gestation	**No villi** (incomplete, already passed, or **ectopic**); **vesicles** (mole); **fat, omentum or bowel** (perforation)
Vitals every 15 min for 30–60 min	Pulse and BP stable	Rising pulse, falling BP: haemorrhage, perforation or haematometra
Bleeding	Light, like a period	Brisk bleeding: re-aspirate, uterotonic, check the cervix
Pain	Cramps settling with analgesia	Severe or increasing pain with an enlarged tender uterus (**haematometra**) or peritonism (**perforation**)
Before discharge	Walking, passing urine, eating	–
Given before she leaves	Anti-D if Rh-negative; contraception; analgesia; danger signs; follow-up in 1–2 weeks	–

Red flags

RED FLAG

- **Cannula passes beyond the expected depth** (e.g. 12 cm in an 8-week uterus) with **loss of resistance**, **sudden severe pain**, or **yellow fat, omentum or bowel** in the aspirate. This is **perforation**: stop at once.
- **Pulse rising and BP falling during or after the procedure.** Think haemorrhage (atony, cervical tear, perforation) or haematometra.
- **Circumoral numbness, metallic taste, tinnitus, twitching, then seizures** during the block. This is **lidocaine toxicity** (intravascular injection or overdose).
- **Sudden bradycardia, pallor and fainting with cervical traction or dilation.** A **vasovagal** reaction.
- **No villi in the aspirate** when you expected them. Consider an **ectopic** or an incomplete procedure.
- **Grape-like vesicles** in the aspirate: a **molar pregnancy**. Send for histology and follow hCG.

7. Clinical Acumen

Thinking like a clinician

- **Know the uterus before you enter it.** Bimanual exam for size and position is the most important safety step. Use tenaculum traction to **straighten the uterocervical angle**; this protects against perforation more than any other manoeuvre.
- **Never force.** If the cannula or dilator meets resistance at the internal os, stop and use a smaller size. Consider **misoprostol 400 µg sublingually 1 hour (or vaginally 3 hours) before** for a closed cervix in missed abortion.
- **When the aspirator stops sucking**, work out why before you blame "completion":
  - The aspirator is **full** (60 mL): empty it and recharge.
  - The **cannula is clogged** with tissue: withdraw it and clear it with forceps.
  - The cannula aperture has **slipped out through the os**: you lose the seal and the vacuum.

    - The valves were not properly **closed or charged**.

- **Septic abortion is a reason to hurry, not to wait**: antibiotics first, then the evacuation is source control.
- **Shock with POC in the os**: removing the tissue with sponge forceps can reverse "cervical shock" before you even start the MVA.

When to abandon and escalate

- suspected perforation;
- bleeding that does not stop with completion and uterotonics;
- a uterus larger than you can safely empty;
- a patient who cannot tolerate it despite adequate analgesia (convert to theatre with sedation);
- suspected mole in a large uterus (move to a hospital with blood and electric suction).

Common student and junior errors

- Skipping the bimanual exam.
- Injecting lidocaine without aspirating first.
- Starting before the block has worked (wait 2–3 minutes).
- Pushing the plunger while the cannula is still attached inside the uterus.
- Not looking at the tissue.
- Leaving the instruments to dry with blood on them.
- Forgetting FP.

Worked examples

Worked example 1: inspecting the aspirate. MVA for "incomplete abortion" at 7 weeks yields only 10 mL of blood clot. Floated in water against a light: **no villi, no sac**.

- Interpretation: either the POC were already passed (complete abortion), the evacuation was incomplete, or **this is an ectopic pregnancy**.
- Next: check vitals and the abdomen, arrange an **ultrasound** (adnexal mass, free fluid) and a **serum hCG**, repeat in 48 h. Do not discharge her without an ectopic safety-net.

Worked example 2: a retroverted uterus. Bimanual before MVA: 9-week-size uterus, **retroverted**. The os is partly open.

- Apply the tenaculum to the **posterior lip** (or anterior with firm traction) to **straighten the uterocervical angle**; direct the cannula **posteriorly** along the axis of the uterus; expect the fundus at about 9–10 cm.
- If the cannula advances beyond the expected depth without resistance: **stop, suspect perforation**, and follow the perforation pathway.

> **UGANDA CONTEXT**
> A study at Mbarara Regional Referral Hospital found that **a large share of women still experience significant pain during MVA**. A study in Northern Uganda found gaps in health workers' MVA knowledge and practice at lower-level facilities. Doing the basics well (a pre-procedure NSAID, a properly placed paracervical block, waiting for it to work, and calm verbal support) is part of quality PAC.

8. Complications and Troubleshooting

Managing complications of MVA

Complication	Recognition	Management
Vasovagal reaction ("cervical shock")	Bradycardia, pallor, sweating, fainting during traction or dilation	Stop traction; lie flat with legs raised; oxygen; IV fluids. **Atropine 0.5 mg IV** if bradycardia persists. Remove any POC in the os.
Incomplete evacuation	Continuing bleeding or cramps; tissue less than expected; later fever or bleeding	**Re-aspirate** (fresh cannula, check the aspirator), or misoprostol 400–600 µg
Haemorrhage	Brisk bleeding during or after the procedure	Check for **retained tissue** (re-aspirate). **Uterine atony:** bimanual massage + **oxytocin 10 IU IM** (or 20 IU in 1 L at 60 drops/min), **misoprostol 400–800 µg**, or **ergometrine 0.2 mg IM** (not in hypertension). **Cervical tear:** pressure or suture. **Perforation or coagulopathy:** see below. Resuscitate, crossmatch.
Uterine perforation	Cannula or dilator goes beyond the expected depth, loss of resistance, sudden pain, fat or bowel in the aspirate	**Stop.** Small fundal perforation, stable patient, nothing abnormal aspirated: **admit**, monitor vitals, Hb and abdomen for 24 hours, give **antibiotics** and a **uterotonic**; complete the evacuation later under ultrasound or laparoscopic guidance by an experienced provider. **Laparotomy** if unstable, bowel or omentum aspirated, peritonitis or expanding haematoma.
Haematometra (post-abortion syndrome)	Minutes to hours later: severe cramping, **enlarged tender uterus**, little external bleeding	**Re-aspirate** (clots), then a uterotonic
Lidocaine toxicity	Perioral numbness, tinnitus, metallic taste, drowsiness, then **seizures**, arrhythmia, arrest	Stop injecting; call for help; **ABC, oxygen**; **diazepam 10 mg IV slowly** (or rectal) for seizures; CPR if arrest; lipid emulsion 20% if available
Infection / endometritis	Fever, offensive discharge, uterine tenderness days later	Antibiotics (e.g. **doxycycline 100 mg orally 12-hourly + metronidazole 400 mg orally 8-hourly for 7 days**, or IV if severe). Re-evacuate if tissue is retained.
Cervical laceration	Bleeding from the tenaculum site	Pressure with sponge forceps, silver nitrate, or a figure-of-eight absorbable suture
Continuing pregnancy / failed procedure	No villi, symptoms continue, rising hCG	Scan; repeat the procedure; exclude ectopic
Air embolism (extremely rare)	Sudden collapse, dyspnoea	Prevention: **never push the plunger while connected to a cannula in the uterus.** Left lateral head-down position, oxygen, CPR.
Missed ectopic	Empty aspirate, persistent pain, positive hCG	Scan; see *Ectopic pregnancy*

Troubleshooting the aspirator

Problem	Cause	Fix
No suction on releasing the valves	Valves not closed or plunger arms not locked; O-ring dry or damaged; liner badly seated	Recharge; lubricate the O-ring; reseat the liner; test the vacuum
Suction stops mid-procedure	Cylinder full; cannula clogged; aperture outside the os	Detach and empty; withdraw and clear the cannula with forceps; keep the aperture inside the os
Plunger springs back	Arms not locked over the rim	Re-lock the arms before releasing the valves
Cannula will not pass the os	Os too narrow; uterus acutely flexed	Tenaculum traction; gentle progressive dilation; smaller cannula; never force

Aftercare checklist

1. **Vitals and bleeding** monitored for 30–60 minutes; she can walk and pass urine; pain is controlled.
2. **Analgesia to take home:** ibuprofen 400 mg 8-hourly or paracetamol 1 g 8-hourly for 2–3 days.
3. **Anti-D** if Rh-negative (within 72 hours).
4. **Contraception:**
   - Any method **immediately** after first-trimester MVA (MEC 1), including an **IUD inserted at the end of the procedure**.
   - **No IUD if septic** (MEC 4).
5. **HIV test and linkage**; syphilis screening; **iron** if anaemic.
6. **Documentation:** indication, uterine size, cannula size, analgesia, estimated blood loss, **tissue findings**, complications, FP method given.
7. **Follow-up:** 1–2 weeks, and immediately with danger signs.

What to do where

Level	MVA-related role
HC III	MVA for uncomplicated incomplete abortion by a trained midwife or clinical officer, under 12 weeks. Refer: sepsis with shock, suspected perforation, uterus over 12 weeks, suspected mole.
HC IV	All of the above plus management of complications (transfusion, laparotomy for perforation), second-trimester PAC with misoprostol, and back-up for HC IIIs
Regional / national referral	Complex cases: large molar pregnancies (electric vacuum aspiration), D&E, bowel injury, ICU care; training and supervision of MVA providers

9. Documentation and Presenting Findings

Model one-line summary (before the procedure):

> "A 22-year-old P1+1 at 10 weeks by dates with an incomplete abortion, **stable** (pulse 92, BP 116/74, afebrile), Hb 10.4 g/dL, Rh-positive, HIV-negative. Os open with POC; **uterus 8-week size, anteverted**, non-tender, adnexae free. **Plan: MVA with an 8 mm cannula** under paracervical block after ibuprofen 800 mg; she has chosen a **copper IUD** after the procedure."

Model procedure note / presentation:

> "MVA done for incomplete abortion at 9 weeks. Uterus 8-week size, anteverted. Ibuprofen 800 mg given 45 minutes before; paracervical block with 1% lidocaine 16 mL. Os open; no dilation needed. **8 mm cannula**; about 60 mL of blood and tissue aspirated. **Villi and decidua seen** on flotation. Completion confirmed by pink foam, gritty sensation and the uterus gripping the cannula. EBL 100 mL; no complications. **Copper IUD inserted.** Vitals stable at 30 minutes. Instruments sent for processing. Review in 2 weeks."

The note must record: indication, uterine size and position, analgesia (drug, dose, time), block volume, dilation, cannula size, volume aspirated, **tissue findings**, signs of completion, estimated blood loss, complications, FP method, anti-D, vitals after, follow-up date.

10. OSCE Checklist

Step	Done well when	Critical?
Greets, introduces self, confirms identity, explains the procedure, **obtains consent**, offers a chaperone	Risks and alternatives stated; FP offered	Yes
Confirms indication, uterine size and position on **bimanual exam**, checks vitals and Hb, asks about allergies	States size and anteverted/retroverted	Yes
Gives **ibuprofen 400–800 mg** 30–60 minutes before; bladder emptied	Timing stated	No
Names and checks the kit: aspirator parts, cannulae, dilators, tenaculum, speculum, sponge forceps, antiseptic, lidocaine	Names each aspirator part	No
Assembles, lubricates, charges and tests the aspirator (valves closed, plunger arms locked)	Hears the rush of air, recharges	Yes
Hand hygiene, PPE, **no-touch technique**	Never touches the insertable part of the cannula	Yes
Inserts the speculum, removes POC from the os, **cleans the cervix** with antiseptic	Spiral outwards, 2–3 times	No
Paracervical block: lidocaine 1% 20 mL or less; 12 o'clock then 4 and 8 o'clock at the cervicovaginal junction; **aspirates before injecting**; waits 2–3 minutes	Dose checked against weight	Yes
Applies the tenaculum with **gentle traction** to straighten the canal	Traction maintained throughout	No
Selects the **appropriate cannula**; dilates gently if needed	Size ≈ weeks; never forces	No
Inserts the cannula to the fundus, then withdraws 1 cm; **attaches the charged aspirator**; releases the valves	Aspirator attached only after insertion	Yes
Rotates and moves the cannula in and out **without withdrawing past the os**	Holds by the cylinder	No
States the **4 signs of completion**	Foam, grit, grip, cramps	No
Closes the valves, **detaches before emptying**; never pushes the plunger into the uterus	–	Yes
Inspects the tissue in water or vinegar against light; states what is expected	Names villi, decidua, sac; knows no villi = ectopic risk	Yes
Removes instruments, checks for bleeding; monitors vitals	Every 15 min for 30–60 min	No
Instrument processing: keep wet or soak in 0.5% chlorine → clean → HLD or sterilise → store	Clean before disinfect	Yes
Post-abortion FP offered; anti-D; danger signs; follow-up; documentation	–	No
Communicates empathetically throughout ("vocal local")	–	No

11. Key Points

High-yield points

- MVA replaces sharp curettage for first-trimester evacuation (less bleeding, pain, perforation, Asherman syndrome).
- **Bimanual for size and position** before any instrument; a missed retroversion causes perforations.
- Septic abortion: **IV antibiotics first, then MVA** (source control). Ectopic or perforation: **do not aspirate**.
- Ibuprofen 400–800 mg 30–60 min before; lidocaine 1% up to 20 mL, **max 4.5 mg/kg**, aspirate before injecting, wait 2–3 min.
- Cannula size ≈ uterine size in weeks; largest that passes the open os in incomplete abortion.
- Completion: **foam, grit, grip, cramps**.
- **Never push the plunger into the uterus.**
- **Always inspect the tissue**: no villi = think ectopic.
- Monitor every 15 min for 30–60 min; anti-D within 72 h if Rh-negative; contraception the same day.
- Processing: **keep wet → clean → HLD/sterilise → store**; 0.5% chlorine = 1 part JIK to 6 parts water.

Numbers & doses

Item	Value
Aspirator capacity	**60 mL**
Vacuum	About **60–66 cm Hg** (24–26 in Hg)
Cannula sizes	**4–12 mm** (4, 5, 6, 7, 8, 9, 10, 12)
Cannula rule	Size in mm ≈ **uterine size in weeks**
Routine upper limit	About **12 weeks'** uterine size (WHO: up to 14 with skill and the right cannula)
Pre-procedure NSAID	**Ibuprofen 400–800 mg** orally 30–60 min before
Lidocaine for block	**1%** (10 mg/mL), up to **20 mL**; max **4.5 mg/kg** plain
Block sites	1–2 mL at 12 o'clock (tenaculum), then **4 and 8 o'clock**, 1.5–3 cm deep, 3–5 mL each
Wait after block	**2–3 minutes**
Cervical priming	Misoprostol **400 µg SL 1 h** or **vaginally 3 h** before
Uterus over 12–14 weeks	Misoprostol **400 µg every 3 hours**
Prophylactic antibiotic (WHO)	Doxycycline 200 mg or metronidazole 500 mg orally, once before
Perforation risk	**Under 1 in 1,000**
Post-procedure monitoring	Every **15 min** for **30–60 min**
Vasovagal bradycardia	**Atropine 0.5 mg IV** if persistent
Lidocaine toxicity seizures	**Diazepam 10 mg IV** slowly (or rectal)
Uterine atony	Oxytocin **10 IU IM** (or 20 IU in 1 L at 60 drops/min); misoprostol **400–800 µg**; ergometrine **0.2 mg IM** (not in hypertension)
Endometritis	Doxycycline **100 mg 12-hourly** + metronidazole **400 mg 8-hourly** orally for **7 days**
Anti-D	Within **72 h** if Rh-negative

Item	Value
HLD	Boil **20 min**; or 0.5% chlorine or 2% glutaraldehyde **20 min**
Autoclave	121 °C, **30 min** wrapped / 20 min unwrapped
0.5% chlorine from JIK 3.5%	**1 part JIK : 6 parts water**

Exam traps

- **"The os is open, so bimanual is unnecessary."** Wrong: you need **size and position** before any instrument goes in; a missed retroversion causes perforations.
- **"Septic abortion is a contraindication to MVA."** Wrong: give IV antibiotics, then MVA is the **source control**.
- **"Push the plunger to clear a blocked cannula."** Wrong: **never push the plunger while connected to a cannula in the uterus** (air embolism). Withdraw and clear with forceps.
- **"The aspirator stopped sucking, so the uterus is empty."** Wrong: check it is not **full, clogged, outside the os, or uncharged**. Completion needs foam, grit, grip and cramps.
- **"20 mL of 1% lidocaine is always safe."** Wrong for a woman under about 45 kg: the maximum is **4.5 mg/kg** (18 mL for 40 kg).
- **"Boil or disinfect the instruments straight after use."** Wrong: **clean first** (detergent and a soft brush); HLD fails on dirty instruments.
- **"No tissue was obtained, so she can go home."** Wrong: no villi = think **ectopic**; scan and serial hCG.

Questions seniors ask

Viva questions

1. *What vacuum does the MVA aspirator produce, and what is its capacity?* About **60–66 cm Hg**; **60 mL**.
2. *Why is MVA preferred to D&C?* Less blood loss, pain, perforation and Asherman syndrome; no general anaesthesia; can be done by midwives at lower-level units; cheaper.
3. *Where do you inject a paracervical block, and with what?* Lidocaine 1%, **up to 20 mL** (maximum 4.5 mg/kg). 1–2 mL at the tenaculum site, then **4 and 8 o'clock** at the cervicovaginal junction, 1.5–3 cm deep, aspirating first.
4. *List the signs that the uterus is empty.* Red or pink foam and no tissue, a gritty sensation, the uterus contracting and gripping the cannula, and increased cramping.
5. *What would make you suspect perforation?* The instrument passes further than expected without resistance, sudden pain, fat or bowel in the aspirate, or haemodynamic deterioration.
6. *What are the steps of instrument processing?* Decontaminate / keep wet → clean → HLD (boil 20 minutes, or 0.5% chlorine or 2% glutaraldehyde for 20 minutes) or sterilise (autoclave or glutaraldehyde 8–10 hours) → store.
7. *How do you make 0.5% chlorine from JIK?* 1 part JIK (3.5%) + 6 parts water.
8. *Can MVA be done in septic abortion?* Yes, **after IV antibiotics are started**. It is source control.

On the ward round

- **What do you check before you pick up the aspirator?** Stable vitals, confirmed indication, ectopic excluded, antibiotics if septic, Hb and group, consent, analgesia given, bimanual size and position, and a charged, tested aspirator.
- **How do you choose the cannula?** About the uterine size in weeks; in incomplete abortion the largest that passes the open os comfortably.
- **How do you know the uterus is empty?** Red or pink foam, no more tissue, a gritty feel, the uterus gripping the cannula, and more cramping.

- **What do you see when you float the aspirate?** Villi (fluffy, frond-like), decidua and the sac; vesicles mean a mole; fat or bowel mean perforation.
- **The cannula goes to 14 cm in a 10-week uterus. What now?** Stop, withdraw, no further suction; monitor vitals and abdomen, antibiotics and a uterotonic; laparotomy if unstable, peritonitic, or fat or bowel was aspirated.
- **She becomes pale with a pulse of 44 as you apply the tenaculum. Why and what do you do?** Vasovagal reaction: stop traction, lie her flat with legs up, oxygen, fluids, remove any POC from the os, **atropine 0.5 mg IV** if bradycardia persists.
- **What must happen before she leaves?** Vitals checked for 30–60 min, anti-D if Rh-negative, contraception, analgesia, danger signs and a 1–2 week review.

Mnemonics

- **Setting up the aspirator: "C-L-O-S-E, then P-U-L-L"**: close the valves, then pull the plunger until the arms **L**ock. Check they lock, or the plunger may spring back.
- **Signs of completion: "F-G-G-C"**: **F**oam (red or pink), **G**ritty feel, **G**rip (uterus contracts on the cannula), **C**ramping increases.
- **Paracervical block: "12, then 4 and 8, aspirate, wait"**.
- **Processing: "Keep wet, Clean, Kill, Keep"** (Soak → Scrub → HLD/Sterilise → Store).
- **Chlorine dilution: "3.5 over 0.5, minus 1 = 6"**.
- **Perforation: "Too deep, too easy, too painful, too fatty"**.

OSCE tips

- Say each step out loud while you do it on the model, including **aspirating before injecting lidocaine**.
- Test and recharge the aspirator in front of the examiner.
- Finish with tissue inspection, FP, anti-D and processing: many candidates stop at "aspiration complete".

> **EXAM FAVOURITE**
> Examiners love the **order of processing** ("clean before you disinfect") and the **"never push the plunger into the uterus"** rule. In the OSCE, say each step out loud while you do it on the model, including aspirating before injecting lidocaine.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Capacity and vacuum of the Ipas-type MVA aspirator	**60 mL** capacity; vacuum of about **60–66 cm Hg (24–26 in Hg)** – similar to an electric pump.
2	Name the parts of the MVA aspirator	Cylinder, plunger with plunger arms, **O-ring**, collar stop (retainer tab), valve with **two buttons**, valve liner and cap.
3	Cannula sizes available in a standard MVA kit	**4, 5, 6, 7, 8, 9, 10 and 12 mm** (colour-coded); dots mark cm from the tip.
4	Rule of thumb for choosing cannula size	Cannula (mm) ≈ **uterine size in weeks**; in incomplete abortion choose the **largest that passes the open os comfortably** (too small clogs, too large tears).
5	Main indications for MVA	**Incomplete/inevitable abortion**, missed abortion, induced abortion (legal indications), septic abortion (after antibiotics), small molar pregnancy, endometrial biopsy – uterine size up to ~12 (–14) weeks.
6	Is septic abortion a contraindication to MVA?	**No.** Start IV antibiotics first, then evacuate promptly – evacuation is source control.
7	Paracervical block: drug, volume, maximum dose	**Lidocaine 1%**, up to **20 mL** (200 mg); maximum **4.5 mg/kg** without adrenaline. Aspirate before every injection.
8	Paracervical block: injection sites	1–2 mL at the **tenaculum site (12 o'clock)**, then **4 and 8 o'clock** (± 2 and 10) at the **cervicovaginal junction**, 1.5–3 cm deep; wait 2–3 minutes.
9	Recommended oral analgesia before MVA	**Ibuprofen 400–800 mg** orally 30–60 min before (± paracetamol 1 g); WHO advises against routine general anaesthesia.
10	Four signs that uterine evacuation is complete	**Red/pink foam** with no more tissue; **gritty sensation**; **uterus contracts/grips** the cannula; **increased cramping**.
11	Why must you never push the plunger while the cannula is in the uterus?	It would push **air and aspirate back into the uterus** (risk of air embolism and infection). Detach first, then empty.
12	Why do you lose vacuum during MVA? (4 causes)	Aspirator **full**; cannula **clogged**; cannula aperture **withdrawn past the os**; valves not properly closed/charged or worn O-ring.
13	Signs of uterine perforation during MVA	Instrument goes **deeper than expected** with **loss of resistance**, **sudden pain**, **fat/omentum/bowel** in aspirate, deteriorating vitals.
14	How do you examine aspirated tissue and what do you look for?	Rinse in a strainer, float in **water or vinegar** in a clear dish **backlit**: **villi**, decidua, **gestational sac**/membranes (fetal parts ≥9 wk). No villi → incomplete, already passed, or **ectopic**; vesicles → **mole**.
15	Sequence of MVA instrument processing	**Keep wet/decontaminate (0.5% chlorine 10 min) → clean (detergent, soft brush) → HLD or sterilisation → dry & store** covered.
16	HLD options for MVA instruments	**Boil 20 min**, or soak in **0.5% chlorine 20 min**, or **2% glutaraldehyde 20 min** (rinse chemicals off with boiled water).
17	Sterilisation options for MVA instruments	**Steam autoclave** (121 °C, 30 min wrapped) for items rated for it (aspirator, metal instruments); **2% glutaraldehyde 8–10 hours**. Many plastic cannulae must NOT be autoclaved.
18	How to make 0.5% chlorine from JIK (3.5%)	Parts water = (3.5 ÷ 0.5) – 1 = **6 → 1 part JIK : 6 parts water**; make fresh daily.

#	Question	Answer
19	What is post-abortion syndrome (haematometra) after MVA?	Blood/clot accumulates in the uterus minutes–hours later: **severe cramps, enlarged tender uterus, little external bleeding**. Treat by **re-aspiration** + uterotonic.
20	Early symptoms of lidocaine toxicity	**Perioral numbness, metallic taste, tinnitus**, light-headedness → twitching → **seizures**, arrhythmia. Stop, ABC, oxygen, diazepam for seizures.
21	Which contraceptive can be inserted at the end of an uncomplicated MVA?	**Copper IUD / LNG-IUS** (MEC 1 after first-trimester abortion); implants and injectables also immediately. Not IUD after septic abortion (MEC 4).
22	Pre-MVA checklist: five things to confirm before the aspirator is used	1. **Indication** (e.g. incomplete abortion under 12 weeks; ectopic excluded) 2) **Stable** or resuscitation running; IV antibiotics already given if septic 3) **Hb and Rhesus group** 4) **Consent** + analgesia given 30–60 min before 5) **Bimanual: size and position**; aspirator charged and vacuum tested.
23	Uterine size at which the fundus becomes just palpable above the symphysis pubis	About **12 weeks**. A uterus clearly palpable abdominally (about **14 weeks or more**) is beyond routine MVA: use misoprostol 400 µg every 3 h.
24	MVA aspirate shows no villi and no sac. What must you consider and do?	Incomplete evacuation, a complete abortion already passed, or an **ectopic pregnancy**. Check vitals and abdomen, **ultrasound** and **serum hCG** (repeat at 48 h); do not discharge without an ectopic safety-net.
25	Post-MVA monitoring: what, how often, for how long?	**Pulse, BP, bleeding and pain every 15 minutes for 30–60 minutes.** Rising pulse or falling BP = haemorrhage, perforation or haematometra.
26	Consent for MVA: the risks to mention	**Bleeding, infection, incomplete evacuation (needing a repeat), cervical injury, uterine perforation (under 1 in 1,000)**, and pain; alternatives are misoprostol or expectant care. Offer contraception.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. You are assisting a midwife at an HC IV to perform MVA for incomplete abortion. After releasing the valves, blood flows briskly into the aspirator, then flow suddenly stops although tissue is still felt in the uterus. The cylinder is about one-third full. What is the MOST likely cause?

A The uterus is empty
B The cannula aperture has been withdrawn past the internal os or is clogged
C The aspirator is full
D Uterine perforation has occurred
E The paracervical block has worn off

2. A 60 kg woman is to have MVA. The clerk draws up 30 mL of 1% lidocaine for the paracervical block. What is the BEST response?

A Proceed – 30 mL is the standard dose
B Use 30 mL but inject quickly to reduce pain
C Switch to 2% lidocaine 20 mL for a denser block
D Reduce to a maximum of 20 mL (200 mg), aspirate before each injection
E Omit the block and give general anaesthesia

3. During MVA for a 10-week incomplete abortion, the cannula suddenly advances to 14 cm without resistance, the woman cries out in pain, and yellow fatty tissue is seen in the aspirate. She is haemodynamically stable. What is the MOST appropriate management?

A Stop, and arrange laparotomy to inspect the bowel and repair the perforation
B Continue aspiration gently to complete the evacuation
C Give oxytocin and discharge home after 2 hours
D Switch to a smaller cannula and continue
E Repeat MVA in one week

4. Which step in processing MVA instruments is MOST important for successful high-level disinfection?

A Soaking in 0.5% chlorine for 10 minutes
B Air-drying before storage
C Boiling for 20 minutes
D Autoclaving at 121 °C
E Thorough cleaning with detergent and water after disassembly

5. A nurse prepares 0.5% chlorine solution from JIK (3.5% sodium hypochlorite). How many parts of water should be added to 1 part of JIK?

A 3
B 5
C 6
D 7
E 10

6. A 26-year-old is 20 minutes post-MVA for missed abortion at 9 weeks. She has severe cramping, pulse 104, and the uterus is palpable at 14-week size and tender, with minimal vaginal bleeding. What is the MOST likely diagnosis and management?

- A Uterine perforation – laparotomy
- B Normal post-procedure cramps – ibuprofen only
- C Haematometra – re-aspirate the uterus and give a uterotonic
- D Septic abortion – IV antibiotics
- E Ectopic pregnancy – laparotomy

7. While injecting a paracervical block, the patient complains of tingling around her lips, a metallic taste and ringing in her ears. What is the MOST appropriate immediate action?

- A Inject the remaining lidocaine quickly and start the procedure
- B Give IV atropine 0.5 mg
- C Give adrenaline 1 mg IV
- D Reassure her that this is anxiety
- E Stop injecting, call for help, give oxygen and monitor; be ready to treat seizures with diazepam

8. A 30-year-old with an incomplete abortion at 8 weeks has temperature 38.9 °C, offensive discharge and a tender uterus. BP 110/70, pulse 108. What is the BEST timing of MVA?

- A After starting IV antibiotics, as soon as possible (within hours)
- B After 48 hours of IV antibiotics when she is afebrile
- C Immediately, before antibiotics, to remove the source
- D MVA is contraindicated; use misoprostol only
- E Only after she has had 7 days of oral antibiotics

9. Which MVA indication is MOST appropriate for use of a 4 mm cannula?

- A Incomplete abortion at 11 weeks' uterine size
- B Missed abortion at 12 weeks
- C Molar pregnancy with a 16-week uterus
- D Endometrial biopsy in a woman with abnormal uterine bleeding
- E Evacuation of retained placenta after term delivery

10. A 24-year-old is booked for MVA at 10 weeks. Bimanual examination is skipped because 'the os is open'. The cannula is pushed anteriorly and perforates. What is the MOST likely preventable factor?

- A Using lidocaine 1% instead of 2%
- B Not identifying a retroverted uterus and not applying tenaculum traction
- C Using ibuprofen before the procedure
- D Choosing a plastic rather than metal cannula
- E Doing the procedure in a treatment room rather than theatre

11. An 8 mm cannula is being used for MVA. Pink foam appears, the uterus grips the cannula and there is a gritty sensation. The aspirate floated in water shows only blood clot and no villi or membranes. The woman had presented with bleeding at 7 weeks and a closed os; no scan was done. What is the MOST appropriate next step? Advanced

- A Discharge – the uterus is empty
- B Repeat MVA with a larger cannula
- C Check serum hCG and arrange ultrasound to exclude ectopic pregnancy
- D Give misoprostol 800 µg
- E Start methotrexate immediately

12. Which statement about the Ipas-type MVA aspirator is CORRECT?

A Emptying is done by pushing the plunger with the cannula still in the uterus
B The O-ring should be left dry to maintain the seal
C The aspirator should be attached to the cannula before inserting the cannula
D The plunger arms must lock over the cylinder rim when charging
E Glutaraldehyde needs only 5 minutes for sterilisation

13. During MVA a woman becomes pale and sweaty, and her pulse falls to 44/min as the tenaculum is applied and the cervix dilated. BP 80/50. What is the MOST appropriate immediate management?

A Continue quickly to finish the procedure
B Laparotomy for suspected perforation
C Give adrenaline 1 mg IM
D Give 20 mL more lidocaine
E Stop traction, lie her flat with legs raised, give oxygen and IV fluids; atropine 0.5 mg IV if bradycardia persists

14. A newly qualified doctor asks which pain-management approach WHO recommends for MVA in the first trimester. Which is BEST?

A General anaesthesia routinely
B Paracervical block plus NSAID, with verbal support; offer conscious sedation where available
C No analgesia is needed as MVA is painless
D IV morphine 10 mg for all patients
E Paracetamol alone

15. Which uterine size is generally the upper limit for MVA in routine practice at an HC III?

A 6 weeks
B 8 weeks
C 12 weeks
D 18 weeks
E 22 weeks

Answers & explanations

> **1. B** · The cannula aperture has been withdrawn past the internal os or is clogged
>
> Loss of vacuum with the cylinder only one-third full suggests **the aperture slipped out through the os (seal lost) or tissue clogged the cannula**. **A**: an empty uterus gives pink foam and a gritty feel, not sudden cessation with tissue remaining. **C**: the cylinder is not full. **D**: perforation is signalled by excessive depth, loss of resistance, pain or abnormal tissue – not simply stopped flow. **E**: analgesia does not affect suction.

> **2. D** · Reduce to a maximum of 20 mL (200 mg), aspirate before each injection
>
> Standard paracervical block is **up to 20 mL of 1% lidocaine (200 mg)**, maximum 4.5 mg/kg plain (270 mg for 60 kg is an absolute ceiling, but 200 mg is the recommended maximum volume). **A/B**: 300 mg exceeds the recommended dose; rapid injection increases toxicity risk. **C**: 2% 20 mL = 400 mg – toxic. **E**: WHO advises against routine GA.

> **3. A** · Stop, and arrange laparotomy to inspect the bowel and repair the perforation
>
> Fat/omentum in the aspirate = **perforation with possible bowel/omental injury → stop and laparotomy** even if stable. **B/D** could cause further visceral injury. **C/E**: observation alone is only acceptable for a small fundal perforation with nothing abnormal aspirated.

4. E · Thorough cleaning with detergent and water after disassembly

Cleaning removes the organic material that shields micro-organisms; HLD or sterilisation **fails on dirty instruments**. **A**: decontamination soak reduces risk to handlers but does not replace cleaning. **C/D** are effective only on clean items. **B** is for storage.

5. C · 6

Parts water = (% concentrate ÷ % desired) – 1 = (3.5 ÷ 0.5) – 1 = **6**. **A/B** give too-strong solutions (corrosive, wasteful). **D**: 7 is the ratio (3.5/0.5) forgetting to subtract 1 – slightly too weak. **E**: much too weak for HLD.

6. C · Haematometra – re-aspirate the uterus and give a uterotonic

An enlarging, tender uterus with little external bleeding shortly after MVA = **haematometra (post-abortion syndrome)**; treat by **re-aspiration** of clot + uterotonic. **A**: no signs of perforation (peritonism, abnormal aspirate). **B**: an enlarged tender uterus is not normal. **D**: 20 minutes is too early for sepsis. **E**: villi were presumably obtained at MVA; the uterus is the problem.

7. E · Stop injecting, call for help, give oxygen and monitor; be ready to treat seizures with diazepam

These are early signs of **lidocaine toxicity** (probable intravascular injection). **Stop, ABC, oxygen, monitor**, treat seizures with **diazepam**; lipid emulsion if available. **A** worsens toxicity. **C**: adrenaline is for arrest/anaphylaxis. **D**: dangerous dismissal. **B**: atropine treats vagal bradycardia, not toxicity.

8. A · After starting IV antibiotics, as soon as possible (within hours)

Septic incomplete abortion: **start IV antibiotics, then evacuate promptly** (source control). **C** increases bacteraemia. **B/E**: delay leaves infected necrotic tissue – linked to maternal deaths in Ugandan audits. **D**: infection is not a contraindication; MVA is faster and more reliable than misoprostol in sepsis.

9. D · Endometrial biopsy in a woman with abnormal uterine bleeding

Small cannulae (e.g. 4 mm) are used for **endometrial biopsy** and very early pregnancy. **A/B** need ~9–12 mm cannulae. **C**: a 16-week mole needs electric vacuum aspiration with larger cannulae at a hospital. **E**: MVA kits are not designed for term retained placenta (manual removal under anaesthesia).

10. B · Not identifying a retroverted uterus and not applying tenaculum traction

Most perforations occur when the **uterine position (retroversion) is unknown** and the **canal is not straightened with tenaculum traction**. **A**: 1% is correct. **C**: NSAIDs are recommended. **D**: plastic cannulae are safer. **E**: MVA is appropriately done in a procedure room.

11. C · Check serum hCG and arrange ultrasound to exclude ectopic pregnancy

No villi from a uterus thought to hold a pregnancy → **complete miscarriage already occurred, or ectopic pregnancy** – check **hCG and ultrasound**. **A** risks missing an ectopic. **B/D**: the uterus is empty by clinical signs. **E**: methotrexate requires a confirmed diagnosis and criteria.

12. D · The plunger arms must lock over the cylinder rim when charging

When charging, **pull the plunger back until the arms lock** over the rim (otherwise it can spring back). **B**: the O-ring is lubricated with silicone. **C**: insert the cannula first, then attach the charged aspirator. **A**: never push the plunger into the uterus. **E**: glutaraldehyde needs 20 min for HLD and 8–10 hours for sterilisation.

13. E · Stop traction, lie her flat with legs raised, give oxygen and IV fluids; atropine 0.5 mg IV if bradycardia persists

Bradycardia and hypotension with cervical manipulation = **vasovagal reaction**. **Stop stimulation, lie flat/legs raised, oxygen, fluids, atropine** if needed. **A** worsens vagal stimulation. **C**: adrenaline is for anaphylaxis/arrest. **D** risks toxicity. **B**: no features of perforation.

14. B · Paracervical block plus NSAID, with verbal support; offer conscious sedation where available

WHO recommends **NSAIDs + paracervical block**, with non-pharmacological support and sedation where available, and **against routine GA**. **A**: unnecessary risk and cost. **C**: MVA is painful – Ugandan data show under-treated pain. **D**: excessive and requires monitoring. **E**: paracetamol alone is less effective than NSAIDs.

15. C · 12 weeks

MVA is routinely used up to about **12 weeks' uterine size** (WHO: vacuum aspiration up to 14 weeks with skilled providers and appropriate cannulae). **A/B**: too restrictive. **D/E**: require misoprostol ± D&E at a hospital.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Routine MVA for incomplete abortion at an HC IV

A 24-year-old P1+1 at 9 weeks by LMP is admitted to an HC IV in Luweero with bleeding and passage of clots. Pulse 92, BP 116/72, temperature 36.8 °C, Hb 10.5 g/dL. Speculum: open os, tissue in the canal. The midwife asks you to help her perform MVA.

1. What must you do before starting the procedure?

> Counsel and **consent**; confirm no sepsis/ectopic features; **bimanual exam: uterine size (8 wk) and position**; group & Rh, HIV test; **ibuprofen 400–800 mg PO 30–60 min before**; empty bladder; prepare, **assemble, lubricate, charge and test the aspirator**; lay out cannulae (7–9 mm), tenaculum, speculum, antiseptic, lidocaine, emergency drugs; discuss FP choice (she wants an IUD).

2. Describe the paracervical block.

> Lidocaine 1%, up to **20 mL**. Inject **1–2 mL at 12 o'clock** and apply tenaculum; with traction, inject **3–5 mL at 4 and 8 o'clock** at the cervicovaginal junction, 1.5–3 cm deep, **aspirating before each injection**; wait **2–3 minutes**, talking to her.

3. Describe the aspiration and how you know you're finished.

> Traction to straighten the canal; insert **8 mm cannula** with rotation to fundus, withdraw 1 cm; attach charged aspirator; release valves; rotate 180° each way and move in and out without leaving the os. **Completion: pink foam, gritty feel, uterus grips cannula, cramping.** Close valves, detach, empty into strainer.

4. What do you do with the tissue and what else before she leaves the couch?

> Float tissue in water/vinegar against light: **villi and decidua** confirm IUP. **Insert copper IUD** (MEC 1). Remove tenaculum, check bleeding. Monitor vitals and bleeding every 15 min for 30–60 min; **anti-D** if Rh-negative; danger signs; follow-up in 2 weeks.

5. What happens to the instruments?

> Immediately **keep wet/soak in 0.5% chlorine 10 min** → disassemble and **clean with detergent and soft brush → HLD (boil 20 min, or 0.5% chlorine/2% glutaraldehyde 20 min, rinse with boiled water)** or sterilise per manufacturer → dry, reassemble, lubricate O-ring, store covered.

Take-away: Safe MVA = know the uterus, block properly, gentle technique, check the tissue, give FP, and process the instruments.

Case 2: Perforation during MVA

At a busy regional referral hospital, an intern performs MVA on a 35-year-old P5 with incomplete abortion after an unsafe procedure at about 11 weeks. The uterus was thought to be 10-week size. The 9 mm cannula suddenly passes to 15 cm; she screams and the aspirate contains yellow lobulated tissue.

1. What has happened and what is your immediate action?

> **Uterine perforation with omentum (or bowel fat) aspirated. Stop the procedure immediately**; do not re-insert blindly; call senior; check vitals.

2. What examination and investigations?

> Pulse, BP, abdominal exam (peritonism, distension), Hb, grouping & crossmatch, ultrasound for free fluid/retained POC, erect CXR if bowel injury suspected, renal function.

3. Outline definitive management.

> Resuscitate (IV lines, fluids); **IV broad-spectrum antibiotics** (ceftriaxone/ampicillin + gentamicin + metronidazole); **uterotonic**; **laparotomy** (omentum/bowel aspirated): inspect bowel, repair uterine defect, evacuate the uterus under direct vision, general surgical team for bowel injury; **tetanus toxoid**.

4. What factors predisposed to this complication?

> Grand multiparity, previous unsafe instrumentation (possible pre-existing injury), soft infected uterus, **uterine position not confirmed** (retroverted?), inadequate traction, operator inexperience, underestimated depth (no attention to cannula dots).

5. What should be documented and communicated?

> Honest disclosure to the patient; clear notes of the event, findings and actions; incident report; audit/learning; PAC follow-up including FP and psychosocial support.

Take-away: Too deep, too easy, too painful, too fatty – stop, never continue blindly, and go to laparotomy when anything other than POC is aspirated.

Case 3: The unprocessed MVA kit

On a Monday morning at an HC III, you find two MVA aspirators left in a dry kidney dish with dried blood since the weekend. The chlorine bucket is cloudy and nobody knows when it was made. A woman with incomplete abortion is waiting.

1. Why is this a problem?

> Dried blood and tissue cannot be reliably removed; HLD/sterilisation **fails on unclean instruments** → risk of transmitting **HIV, hepatitis B/C** and bacteria; the aspirators may also no longer hold vacuum.

2. How do you prepare fresh 0.5% chlorine from JIK?

> Parts water = (3.5 ÷ 0.5) – 1 = 6 → **1 part JIK to 6 parts clean water**; label with date/time; replace daily or when cloudy/bloody.

3. Describe how to reprocess these aspirators.

> Soak to soften (0.5% chlorine 10 min), **disassemble** (valve, liner, plunger, O-ring), **scrub with detergent and soft brush**, rinse, inspect; **HLD by boiling 20 min or chlorine/glutaraldehyde 20 min** (rinse with boiled water), or sterilise; air dry; reassemble; **lubricate O-ring; test vacuum**. Discard if cracked or not holding vacuum.

4. The patient is stable; there is no processed kit available. What are your options?

> **Misoprostol 600 µg PO or 400 µg SL** (uterine size <14 wk, stable, no sepsis) with review in 7–14 days; or refer to the HC IV if heavy bleeding/sepsis. Do not use an unprocessed kit.

5. How would you prevent recurrence?

> Written processing SOP on the wall; assign responsibility per shift; daily chlorine preparation log; enough kits for rotation; IPC training and supervision; include in facility quality-improvement audit.

Take-away: Instrument processing is part of the procedure – clean before you disinfect, and never let

blood dry on the kit.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.